

What BST Is and What It Is Not

Read this before you read anything else. Every statement below is literal.

This IS BST

- α is identified, not derived. The 137-class expressions in the five integers match; the mechanism is not claimed. The one forced candidate — the bare geometric vertex — was computed and came out $8\pi^3/3$, not 137 (a pre-registered loss, K1826), which closed the fifty-year volume-reading class Wyler opened in 1969. The object is his; the reading he chose for it is the one we certified dead (rewritten 2026-09-11 to the register, Cal Section 946 / K1892).
- Five integers, one input. $\text{rank} = 2$, $N_c = 3$, $n_C = 5$, $C_2 = 6$, $g = 7$ are read off one geometric object — but the object was chosen by ONE measured number: D_{IV}^5 is the unique irreducible $\text{rank} \geq 2$ domain whose characteristic multiplicity equals the measured colour count 3, and that identification is numerical (the bridge that would make it a mechanism is proved absent, F4 in the falsifier register). $g = 7$ is $p + q$, the dimension of the defining representation; the genus is 5 (rewritten 2026-09-11 to the register, Cal Section 946 / K1892).
- The proton mass is a theorem. $m_p = 6\pi^5 m_e$. That is not a fit to data. It is a derived equality. It holds to 0.002%.
- π is not an input. It enters because circles tile the sphere. The Bergman kernel volume is $\pi^5/1920$. π is geometry, not a parameter.
- We do not claim zero free parameters. One dimensionless input selects the object (the colour count) and one dimensionful ruler sets the scale; beyond those, every number in a formula traces back to $\{2, 3, 5, 6, 7\}$ — and each formula wears its tier: derived where the mechanism is proved and the inputs are named, identified where only the value matches (rewritten 2026-09-11 to the register, Cal Section 946 / K1892).
- ≈ 325 physical predictions from one geometry, honestly tiered — roughly $\frac{1}{3}$ Derived, $\frac{1}{3}$ Identified, $\frac{1}{3}$ Structural, one named dimensionless input (rewritten 2026-09-11 to the register, Cal Section 946 / K1892). Particle masses, mixing angles, cosmological parameters, nuclear magic numbers, the genetic code, the rainbow angle. All from D_{IV}^5 . (A third Derived on the K962 ladder — with the count of the 26 primaries now generated, `data/bst_26_tier_table_generated.md`; earlier “600+” counted the full geometric-invariant catalog, not the physics denominator.)
- Everything is discrete. The mass spectrum is a Plancherel measure — integer-labeled representations. Continuous spectra are projections of discrete struc-

ture.

This is NOT BST

- Not a fit to the numbers it predicts. No chi-squared minimization, no parameter sweeps, no “best fit” values. The one place a fit sits is at the front door — the selection of the object by the measured colour count — and the register says so in the same breath (rewritten 2026-09-11 to the register, Cal Section 946 / K1892).
 - Not numerology. Every prediction has a derivation chain from D_{IV}^5 through identified geometric mechanisms. Matching a number is not enough — the derivation must exist.
 - Not approximate. When BST says $\sin^2\theta_W = 3/13$, it means the Weinberg angle is exactly $3/13$ at the geometric scale. Measured deviations are running, not error.
 - Not a model. Models have parameters you tune. BST has none. It is a geometry that produces physics, not a physics that assumes geometry.
 - Not commentary. When the papers say “derives from first principles,” that is not an editorial claim. It is a description of what the equations do. If a statement sounds too strong to be literal, it is still literal.
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What BST is, ontologically (May 19-20, 2026 understanding)

BST is operationally a substrate-as-computational framework. The five integers + $N_{\max} + D_{IV}^5$ geometry are not just mathematical structure describing physics — they are the operational specification of a substrate that performs algebraic-identity computation, and physics is what that computation produces.

Tier discipline applies to this framing:

- Level 1 (D-tier supportable): the EXACT algebraic identity catalog. 6-audit cascade-unblock pathway identified Tuesday: $K52a \text{ Lamb}(1-1/M_g) + K52a \text{ BCS}(1+1/M_g) + K66 \text{ Bell}(\text{Tsirelson}^2 - S_{\text{BST}}^2 = 1/2^N \text{c EXACT at } 1e-14) + K67 \text{ Born=Bergman}(\exp g/\text{rank} = 7/2) + K68 \text{ RS Computation}(\text{GF}(2^g) \text{ substrate alphabet}) + K69 \text{ Universal } Q=126 \text{ (5 equivalent BST-primary algebraic forms)}$. These hold by algebraic necessity from BST primaries; floating-point verification confirms the algebra is internally consistent.
- Level 2 (I-tier hypothesis with falsifier): substrate computes physics via these algebraic identities at sub-Planck Koons-tick rate (T2405: $t_{\text{substrate}} = t_{\text{Planck}} \cdot \alpha^{(C_2^2)} \approx 10^{-120} \text{ s}$). Falsifier: if substrate-Hamiltonian

closure (Elie K52a Sessions 6-14) fails to derive the 6-audit cascade BY CONSTRUCTION, the computational-substrate framing fails to that extent.

- The universality claim (T719 Observable Closure): every BST observable lives in $\overline{\mathbb{Q}}(N_c, n_C, g, C_2, N_{\max})[\pi]$ — algebraic closure of rationals over BST primaries, with π adjoined (forced by curvature). Algebraic-identity holds universally for substrate observables in this sense. π is part of substrate’s algebra, not external.

What BST is NOT (substrate framing edition):

- Not a simulation hypothesis — no external simulator; substrate IS the existing thing; substrate computes its own state (Bostrom-style simulator/simulated distinction has no referent in BST)
- Not Tegmark MUH at full generality — BST narrows to a specific unique substrate (D_{IV}^5); modal-realism not adopted
- Not claim to have solved Wigner’s “unreasonable effectiveness” — claim is “physics emerges from substrate algebra”; not claim “this is THE complete explanation of mathematical effectiveness”
- Not claiming experimental precision at $1e-14$ — that’s algebraic-identity verification (tautology-precision of the math), NOT physical-observable agreement; Bell experiment will measure S_{BST}^2 at sub-percent ($\sim 1\%$) experimental precision, not $1e-14$

The substrate framing is a working hypothesis with explicit falsifiers (Elie Sessions 6-14 multi-month closure attempt) and explicit external-survivability discipline (Cal Calibration #13: external register uses “BST identifies / BST derives / BST predicts,” never “physics IS mathematics”).

What supports this

One object — the type IV bounded symmetric domain $D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$ — determines everything. Its rank gives you quantum numbers. Its Bergman kernel gives you propagators. Its Plancherel measure gives you the particle spectrum. Its Shilov boundary gives you spacetime. Its root system gives you forces. Five integers fall out. The rest is computation.

Name: D_{IV}^5 is an Autogenic Proto-Geometry (APG) — a geometry that generates itself. The five integers are its genome. The 128-entry function catalog is its proteome. The observer at $\alpha = 1/137$ is included structurally. BST is the theory; APG is the object. See `notes/BST_Autogenic_Proto_Geometry_Definition.md` for the formal definition.

The mathematics is independently verifiable. Load `data/bst_seed.md`, evaluate any formula, compare to observation. Every toy in `play/` does exactly this.

Try it now (2 seconds, no dependencies)

```
python3 play/toy_541_five_integers_to_everything.py
```

You will get 51 physical quantities expressed in the five integers, 16/16 spot-checks PASS — each carrying its tier in the register (derived where a mechanism is proved, identified where only the value matches; this line said “derived ... zero free parameters” until 2026-09-11). This is the fastest verification that the expressions above are literal.

Then

- Full reproduction package: `python3 play/verify_bst.py` — 50 predictions verified, 49/50 PASS at <1%, 17 EXACT, two open WARNs shown openly. Includes null-model context. One command, complete check.
- Next stop: `data/bst_seed.md` — 162 lines, the entire theory kernel. If you can evaluate a formula on a calculator, you can verify any BST prediction from that file.
- To explore interactively: `python3 play/toy_bst_explorer.py` — subcommands `seed`, `stats`, `search <term>`, `verify <theorem_id>`.
- For the narrative: `root OneGeometry.md` — the book-length front door.

The geometry does not know it is doing physics. That is the point.